claude-windows / fa4dc4eb-4705-47fd-b044-4f017b586b15

Metadata

- Source: `claude-windows`
- Kind: `claude`
- Source file: `C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\fa4dc4eb-4705-47fd-b044-4f017b586b15\subagents\workflows\wf_ac0cbb29-6bf\agent-a595885fdc3994f31.jsonl`
- SHA-256: `9f08e57fccd8ce3f5e7d6ef48ed312f1364da3a20935980fcee92ee26acecec3`
- Source modified: `2026-06-19T15:53:43+00:00`
- Imported at: `2026-07-05T16:48:22+00:00`
- project: `wf_ac0cbb29-6bf`
- session_id: `fa4dc4eb-4705-47fd-b044-4f017b586b15`

Transcript

1. user (2026-06-19T15:52:06.166Z)

```
You are auditing a badminton rally-detection codebase for DOC/COMMENT STALENESS ? statements
that CONTRADICT the current code or data.
HARD RULES:
- Report ONLY factual staleness with CODE EVIDENCE (file:line or symbol) that proves it. Open
the cited code and confirm before reporting.
- NEVER invent a fact or a status. If a thing is built-but-default-OFF / unwired / unmeasured,
say EXACTLY that ? do not overclaim "in production".
- A "[design]" tag is stale ONLY if the thing is actually BUILT in code (cite the symbol).
- DO NOT propose subjective rewrites, style/tone changes, or improvements. DO NOT relitigate
owner-gated DECISIONS (owned-model roadmap, licensing/MIT-Apache stance, product/strategy
choices, genuinely-still-design proposals).
- Preserve the document's voice and formatting; make the SMALLEST surgical edit. stale_text MUST
be copied verbatim with enough context to be unique.
- IGNORE docs/archives/** (terminal snapshots) and any memory/ file.
Working dir is the repo root; use Read/Grep/Glob freely.

ADVERSARIALLY VERIFY these proposed staleness fixes for ONE file:
docs/RALLY_DETECTION_GAP_CLOSURE.md.
For EACH proposed fix: (1) open docs/RALLY_DETECTION_GAP_CLOSURE.md and confirm stale_text
appears VERBATIM (tighten/expand it to an exact, UNIQUE match if needed ? this text will be
applied by exact string match, so it must be perfect); (2) re-open the cited code yourself and
confirm the claim is genuinely contradicted by the CURRENT code; (3) confirm corrected_text is
accurate, preserves voice, and is minimal; (4) DROP it if it is subjective, an owner-gated
decision, already-correct, or you cannot verify the evidence. Output ONLY the survivors (verdict
confirm or revise), with stale_text guaranteed to be an exact unique substring of the file.

PROPOSED FIXES:
[
  {
    "file": "docs/RALLY_DETECTION_GAP_CLOSURE.md",
    "stale_text": "  - *Shuttle-in-hand / out-of-play* ? `shuttle_player_colocation` (shuttle
box ? person box = held) `[design]` from the fusion plan §4,",
    "corrected_text": "  - *Shuttle-in-hand / out-of-play* ? `shuttle_player_colocation`
(shuttle box ? person box = held) `[verified]` built in `fusion_features.py` (default-OFF,
unmeasured on golden) from the fusion plan §4,",
    "why_stale": "Tags shuttle_player_colocation `[design]` (proposed), but the feature is built
in backend/pipeline/detectors/fusion_features.py:168 (in PERSON_FEATURE_NAMES, computed by
compute_person_features) and wired in fusion_hybrid.py under cue_flags.person. Per this doc's
own honesty key, `[design]` is only correct for proposed-not-built; the feature is built but
default-OFF and unmeasured on golden (the generalized rally-END rule that consumes it is what
remains design).",
    "severity": "high"
  },
  {
```

    "file": "docs/RALLY_DETECTION_GAP_CLOSURE.md",
    "stale_text": "person producer in `yolo_hybrid` (built) · `shuttle_player_colocation` (new feature) · `scratch/ingest_golden.py`",
    "corrected_text": "person producer in `yolo_hybrid` (built) · `shuttle_player_colocation` (built in `fusion_features.py`, default-OFF + unmeasured on golden) · `scratch/ingest_golden.py`",
    "why_stale": "The reuse map labels shuttle_player_colocation \"(new feature)\" in a \"what's new vs there\" list, implying not built. It is built in backend/pipeline/detectors/fusion_features.py:168 (PERSON_FEATURE_NAMES line 197) and persisted by fusion_hybrid.py::_person_features. What is genuinely undone is extracting/measuring it on the golden clips (this doc's §7 row P3), not building the feature.",
    "severity": "medium"
  },
  {
    "file": "docs/RALLY_DETECTION_GAP_CLOSURE.md",
    "stale_text": "**Caveat:** two golden clips so far (n=13 corpus, GX010141+GX010137) ? directional, not conclusive.",
    "corrected_text": "**Caveat:** two golden clips so far (n=15 corpus, GX010141+GX010137) ? directional, not conclusive.",
    "why_stale": "The parenthetical states the current corpus size as 'n=13 corpus', but output/human_lovo_manifest.json now has 15 entries. The 'two golden clips so far' (the n=2 ground-truth-scored finding) is left intact; only the current-corpus-size token is corrected.",
    "severity": "medium"
  },
  {
    "file": "docs/RALLY_DETECTION_GAP_CLOSURE.md",
    "stale_text": "- **n is small.** One golden clip scored so far (GX010141); the corpus is n=12 with several weak clips. Per-video conclusions firm up only as the owner labels more (GX010137 / GX020140 / a large-count clip next).",
    "corrected_text": "- **n is small.** One golden clip scored so far (GX010141); the corpus is n=15 with several weak clips. Per-video conclusions firm up only as the owner labels more (GX010137 / GX020140 / a large-count clip next).",
    "why_stale": "Present-tense statement 'the corpus is n=12', but output/human_lovo_manifest.json now has 15 entries. Only the current-corpus-size token is corrected; the 'one golden clip scored so far' progress wording is left to the doc's own §3 which already supersedes it.",
    "severity": "medium"
  },
  {
    "file": "docs/RALLY_DETECTION_GAP_CLOSURE.md",
    "stale_text": "| D3 | **Golden labels = ground truth; Gemini = reference, not truth** | n=12 corpus | Gemini under-detects too ? agreement ? accuracy. Report both, gate on golden. |",
    "corrected_text": "| D3 | **Golden labels = ground truth; Gemini = reference, not truth** | n=15 corpus | Gemini under-detects too ? agreement ? accuracy. Report both, gate on golden. |",
    "why_stale": "The decision-source column characterizes the current corpus as 'n=12 corpus', but output/human_lovo_manifest.json now has 15 entries. The decision itself is unchanged; only the stale corpus-size figure is corrected.",
    "severity": "low"
  }
]

2. assistant (2026-06-19T15:52:08.633Z)

I'll verify each proposed fix by confirming the stale_text appears verbatim in the doc, then checking the cited code.

3. user (2026-06-19T15:52:09.599Z)

Rally Detection ? Closing the Accuracy Gap (golden-eval-driven, per-video)

> **Status:** `DRAFT` ? implementation **owner-gated** · **Owner:** Avi · **Created:** 2026-06-18 · **Last updated:** 2026-06-18
> **Lifecycle:** `DRAFT ? IN PROGRESS ? DONE ? archived` (move to `docs/archives/` when DONE)
> **Tracking anchors:** §7 progress tracker is the source of truth; pointer in memory

`current-state`; add to `docs/README.md` + `docs/NEXT_STEPS.md` when work starts.

> **Relation to existing docs:** **peer-of** [`RALLY_QUALITY_RESEARCH.md`](RALLY_QUALITY_RESEARCH.md) (that is the research + eval-framework doc; this is the *execution tracker* for the remaining accuracy gap). **Builds on** [`MULTI_SIGNAL_FUSION_PLAN.md`](MULTI_SIGNAL_FUSION_PLAN.md) (signal mechanics), [`QUALITY_ITERATIONS.md`](QUALITY_ITERATIONS.md) (the R-log / empirical backbone), [`GOLDEN_VIDEOS.md`](GOLDEN_VIDEOS.md) (the corpus). Does **not** duplicate them ? it points.

> **Honesty note:** each claim tagged `[verified]` (checked vs code/measured) / `[design]` (proposed). The headline numbers below are real measurements, but `n` is small ? read §6.

---

0. TL;DR

We are using the **growing golden corpus** (human rally labels) to find and close the rally-detection accuracy gap **per video, before launch** ? and the first ground-truth clip already reshaped the picture. The boundary work (R2/R3/R4) **holds** ? when local fires on a real rally, its edges are tight (|?| < 1 s). **The remaining gap is DETECTION, not boundaries:** (G1) **recall** ? both local and Gemini miss ~half the rallies on hard clips; (G2) **precision** ? local over-fires (false positives), especially on continuous drilling; (G3) **rally-END over-extension** ? when a player *quickly knocks the shuttle back after a point*, shuttle-velocity alone can't tell "rally" from "casual reset," so the window runs long / merges. n=2 sharpens the priority: **LOCAL = high recall / low precision, GEMINI = high precision / low recall** (opposite, complementary ? on the normal clip local's recall *beats* Gemini's). So **the #1 lever is precision** (cut local's false positives via `rally_gate` Gate-A + colocation; the player-state rally-END rule feeds this), with **signal fusion** (local's recall + Gemini-style precision) as the durable win. **Caveat:** two golden clips so far (n=13 corpus, GX010141+GX010137) ? directional, not conclusive.

1. Problem & goal

The detector is good enough to ship a highlights product, but the golden eval shows clip-dependent accuracy that we want to **close consistently on every video, not just on average** ? catching the failure modes in the quality loop instead of after launch.

- **Concrete target:** on every corpus video, local is **at-par-or-better than Gemini vs the golden labels**, with **no per-video regression** on either eval set (the dual-eval-set discipline, [[eval-dual-set-regression]]).
- **"Good" =** the three gaps below closed enough that a milestone flip (default-ON) clears the over-seg guard AND improves per-video F1 broadly, recorded in a Launch Report ([[launch-report-convention]]).
- **Read first to get current:** [`RALLY_QUALITY_RESEARCH.md`](RALLY_QUALITY_RESEARCH.md) §1?2 (the gap + diagnosis), [`MULTI_SIGNAL_FUSION_PLAN.md`](MULTI_SIGNAL_FUSION_PLAN.md) §4 (in-flight vs in-hand), [`HOW_RALLY_DETECTION_WORKS.md`](HOW_RALLY_DETECTION_WORKS.md) (the 2-layer model).

2. Decisions locked

#	Decision	Source / date	Implication
D1	The gap is **DETECTION (recall + precision + rally-END), not boundary precision**	golden GX010141, 2026-06-18 `[verified]`	Don't build a generic boundary head; spend on rally-state signals.
D2	Measure **per-video AND aggregate; no regression on ANY video**	[[eval-dual-set-regression]] / [[decision-rigor-norm]]	A win must broadly help, not trade one clip for another.
D3	**Golden labels = ground truth; Gemini = reference, not truth**	n=12 corpus	Gemini under-detects too ? agreement ? accuracy. Report both, gate on golden.
D4	New signals land **default-OFF + flag-gated**, promoted only on measured lift	[`EXPERIMENT_HARNESS.md`](EXPERIMENT_HARNESS.md) / [[weak-signals-retained]]	Zero-regression rollout; retained even on a small-n null.
D5	**Rally-END needs a PLAYER-state signal** ? shuttle velocity can't separate a competitive rally from a casual post-point return	owner insight + R4 limitation, 2026-06-18 `[design]`	Add person kinematics / shuttle-in-hand as the rally-end discriminator.

3. Foundation ? the measured gaps (golden ground truth)

Ground-truth-scored Boxhill clips (more land as the owner labels them). Milestone

windowing = cap6 + R4 ([`QUALITY_ITERATIONS.md`](QUALITY_ITERATIONS.md) R3/R4). Local & Gemini vs truth, n=2 so far: `[verified]`
32
33 | clip (golden) | system | found | tIoU@0.5 P/R/F1 | tolF1 | hits | false+ | missed | \|?start\| | \|?end\| |
34 |---|---|---|---|---|---|---|---|---|---|
35 | **GX010141** (29, hard/drilling) | LOCAL | 22 | 0.59/0.45/**0.51** | 0.43 | 13 | 11 | 16 | 0.78 | 0.68 |
36 | | GEMINI | 16 | 0.94/0.52/**0.67** | 0.53 | 15 | 4 | 14 | 0.31 | 0.46 |
37 | **GX010137** (34, normal match) | LOCAL | 39 | 0.77/**0.88**/**0.82** | 0.68 | 30 | 14 | 9 | 0.39 | 0.44 |
38 | | GEMINI | 27 | 1.00/0.79/**0.89** | 0.89 | 27 | 0 | 7 | 0.50 | 0.52 |
39
40 - **Corpus aggregate (n=13, milestone):** tIoU F1@0.5 **0.474** [0.359, 0.579], F1@0.25 0.633; R2 tolF1 0.359. GX010137 (0.822) is the corpus's **best** clip; GX010141 (0.510) is above the mean. `[verified]`
41 - **Opposite failure modes ? the central finding (n=2).** **LOCAL = high recall, low precision; GEMINI = high precision, low recall.** On the normal clip GX010137, local's **recall 0.88 BEATS Gemini's 0.79** (local finds *more* real rallies) but local fires 14 false positives; Gemini fires **zero** false positives but misses 7 rallies. They are **complementary** (on GX010141 they overlap on only ~3 rallies yet each hits ~13?15 of the golden set). ? **fusion (local's recall + Gemini-style precision filtering) is the clear win.**
42 - **The gaps, re-prioritized by the data:**
43 - **G2 ? precision is LOCAL's #1 gap** (over-fires on both clips: 11/22 and 14/39 false positives). This is the cheapest + highest-leverage fix (Gate-A / colocation). Filter local's FPs on GX010137 and it lands ~0.88+ ? at-par with Gemini, since local already out-recalls it.
44 - **G1 ? recall** is clip-dependent: a local *strength* on normal play (? Gemini), but on hard continuous-drilling clips *both* miss ~half (GX010141 recall ~45?52%). The recall lever matters mainly for the hard clips.
45 - **G3 ? rally-END over-extension** (quick-return-after-point): shuttle keeps moving ? window runs long / merges; R4 (shuttle-only) can't catch it. Contributes to G2's false positives/merges.
46 - **Local's gap to Gemini SHRINKS on normal play:** hard clip ??0.16 (0.51 vs 0.67) ? normal clip ??0.07 (0.82 vs 0.89), and local *out-recalls* Gemini there. The deficit is precision, not detection.
47 - **What is NOT the gap:** boundary precision. When local hits a rally its edges are tight (|?start|/|?end| ? ~0.8 s, on GX010137 even tighter than Gemini). The R2/R3/R4 windowing is doing its job. `[verified]`
48 - **Long-rally lens refinement (P7, n=15, `[verified]`):** filtering the eval to >5s rallies (the eye-catching highlights) shows local's single-court over-fire is **short-FP-dominated** (seg_ratio 1.43?1.04 ? duration-filterable), but **multicourt still over-fires 1.58× on LONG rallies** (adjacent-court games are genuine long rallies). ? the multicourt precision gap needs **court-localization (Gate-B / court mask, Lever C)**, NOT a duration filter; and the strict tolerance precision is gated by **over-merge contamination** of the long-pred set, not just short FPs. See [`QUALITY_ITERATIONS.md`](QUALITY_ITERATIONS.md) "P7".
49 - ? **Data caveat:** GX010141 was labeled before ~3 rally-annotator-tool bugs were found + fixed (GX010137 is post-fix). GX010141's numbers may shift on a re-export ? re-verify before drawing hard conclusions from it.
50
51 ## 4. Design / architecture ? the levers (REUSE MAP)
52 The signal mechanics live in [`MULTI_SIGNAL_FUSION_PLAN.md`](MULTI_SIGNAL_FUSION_PLAN.md); this is which lever attacks which gap.
53
54 - **Lever A ? player-state rally-END rule** (targets **G3 + G2**) `[design]`. Generalize R4 from shuttle-only to **shuttle OR bilateral player motion**: end the rally at the last frame with competitive evidence from *either*, and **hard-terminate on a "shuttle picked up" event**. Two cheap detectors, no new model required:
55 - *Player kinematics* ? person velocity drops / players go stationary after a point (the casual returner taps it while the other walks). **Reuse:** the person cue producer (torchvision + ByteTrack) in `yolo_hybrid` (fusion P1, `[verified]` exists).
56 - *Shuttle-in-hand / out-of-play* ? `shuttle_player_colocation` (shuttle box ? person box = held) `[design]` from the fusion plan §4, **or** the no-new-model proxy: **sustained WASB `Visibility=0`** after the last fast stroke (held shuttle goes dark). Use it as a **termination

anchor + reverse-timeline search** to the true end (last net-crossing / last fast-shuttle / last bilateral motion) ? the owner's idea, refined: the hold event is *late*, so anchor-then-backtrack, don't use its timestamp as the end.
- **Lever B ? recall** (targets **G1**) `[design]`. Both systems miss real rallies ? detector sensitivity: proposal-windowing / velocity-threshold tuning, and the **complementary-fusion** angle below. Measure recall floor per video; the highest-leverage labels are the *missed* rallies (active learning, RALLY_QUALITY_RESEARCH §4.3).
- **Lever C ? signal/ensemble fusion** (targets **G1**) `[design]`. Local & Gemini (and shuttle vs person cues) hit *different* rallies ? a union/ensemble lifts recall. Already-built scaffolding: `rally_gate.py` **Gate A** (net-crossings, structural rally-state, `[verified]` built, eval-only ? wire to prod = cheapest precision win for G2) and the FusionSegmenter feature path.
- **Pure-precision cleanup** (**G2**): `rally_gate` Gate A kills single-feed / held-shuttle *starts* structurally (?0?1 net crossings vs a real rally's several) ? the cheapest G2 win and independent of the person stack.

Reuse map (what's new vs there): `rally_gate.py::gate_windows` (Gate A, built) · `tracknet_runner` R4 end-trim (extend, don't replace) · `backend/eval/windowing.py` (preset knobs) · person producer in `yolo_hybrid` (built) · `shuttle_player_colocation` (new feature) · `scratch/ingest_golden.py` + `scratch/boxhill17_score_golden.py` (the per-video golden scoring loop, `[verified]`).

6. Honest limits ? what this does NOT do
- **n is small.** One golden clip scored so far (GX010141); the corpus is n=12 with several weak clips. Per-video conclusions firm up only as the owner labels more (GX010137 / GX020140 / a large-count clip next).
- **Nothing here is shipped or even started** ? implementation is owner-gated. A DONE row in §7 means *measured on golden*, not *flipped on by default*.
- **Player-state levers need GPU person-detection on the golden clips** (not yet extracted); until then Lever A/B are design, not data.
- The boundary tightness is from clips where local *hits* the rally; it says nothing about the rallies it misses (G1).

7. Deliverables & progress tracker ? **source of truth**
Legend: ? Todo · ? In progress · ? Done · ? Gated. One small PR per row, off `origin/master`, no stacking.

ID	Deliverable	Depends on	Gated?	Status	PR
P0	Golden scoring loop (`ingest_golden.py` + `boxhill17_score_golden.py`); GX010141 ingested (#12) + scored	?	No	?	this work
P1	Ingest the rest of the Boxhill golden batch (137/020140 + a large-count clip) ? per-video baseline table	owner labels	? owner	?	?
P2	Wire `rally_gate` **Gate A** (net-crossings) to the served path ? cheapest G2 win	?	? owner	?	?
P3	Extract person-velocity + `shuttle_player_colocation` on the golden clips (default-OFF features)	P1	? owner	?	?
P4	**Lever A**: player-state rally-END rule (R-next) ? measure ?F1 / ?merge-rate / ?|?end| vs R4	P3	? owner	?	?
P5	**Lever B/C**: recall + complementary fusion ? measure recall lift, no precision regression	P3	? owner	?	?
P6	Launch decision: milestone + best levers default-ON, with a Launch Report (both rulers + over-seg guard)	P4,P5	? owner	?	?
P7	**Long-rally (>5s) quality lens** (eval `--min-duration`/`--stratify`) + reel knob (`stitching.min_rally_duration`), both default-OFF. **Measured n=15:** single-court over-fire is short-FP-dominated (seg_ratio 1.43?1.04) but **multicourt over-fires 1.58× even on long rallies** (? court-localization, Lever C) and strict tolP doesn't rise (over-merge-contaminated). See [`QUALITY_ITERATIONS.md`](QUALITY_ITERATIONS.md) "P7".	?	No (eval+cfg, default-OFF)	? measured	this PR

8. Open questions ? owner / external
1. **Which lever first** once labels land ? the no-model wins (Gate A wiring + shuttle-invisibility end-anchor), or go straight to person-velocity? *(decides: owner, after the per-video baseline)*
2. **Per-video target** ? "at-par-with-Gemini-vs-golden on every clip," or an absolute

F1 floor? *(decides: owner)*
3. **Recall lever** ? detector-sensitivity tuning vs ensemble fusion as the primary path for G1. *(decides: data, post-P3)*

9. Definition of done
- [ ] Golden corpus covers the Boxhill set + enough variety that per-video reads are trustworthy (n meaningfully > 12).
- [ ] The **rally-END quick-return** failure mode (G3) is closed: measured ? merge-rate + ? |?end| on the clips that exhibit it, no regression elsewhere.
- [ ] Local is **at-par-or-better than Gemini vs golden on every corpus video** (no per-video F1 regression).
- [ ] The winning levers are flipped **default-ON** via a Launch Report (over-seg guard clean on both rulers).
- [ ] ? flip `Status ? DONE`, archive to `docs/archives/`.

10. References
Code anchors `[verified]`: `backend/pipeline/detectors/rally_gate.py` (Gate A) · `backend/pipeline/detectors/tracknet_runner.py` (R4 end-trim, `_trim_inactive_tail`) · `backend/eval/windowing.py` · `backend/eval/rally_seg_eval.py` + `tolerance_metrics.py` (R2) · `backend/pipeline/segmenters/yolo_hybrid.py` (person cue) · `scratch/ingest_golden.py`, `scratch/boxhill17_score_golden.py`.
Internal docs: [`RALLY_QUALITY_RESEARCH.md`](RALLY_QUALITY_RESEARCH.md), [`MULTI_SIGNAL_FUSION_PLAN.md`](MULTI_SIGNAL_FUSION_PLAN.md) §4, [`QUALITY_ITERATIONS.md`](QUALITY_ITERATIONS.md), [`EXPERIMENT_HARNESS.md`](EXPERIMENT_HARNESS.md), [`GOLDEN_VIDEOS.md`](GOLDEN_VIDEOS.md), [`HOW_RALLY_DETECTION_WORKS.md`](HOW_RALLY_DETECTION_WORKS.md).

Changelog
- 2026-06-18 ? Created. Seeded with the GX010141 golden scoring (first ground-truth clip), the three named gaps (recall / precision / rally-END), and the player-state end-rule design from the owner's quick-return insight.
- 2026-06-18 ? Added GX010137 (golden #13, normal match play): local F1@0.5 **0.82** (recall **0.88 > Gemini 0.79**) vs Gemini 0.89. Confirms **opposite failure modes** (local high-recall/low-precision, Gemini the reverse) ? **precision is local's #1 lever**, fusion the durable win. Flagged the GX010141 pre-annotator-fix data caveat.
- 2026-06-18 ? **P7 long-rally (>5s) quality lens + reel knob (both default-OFF), measured n=15.** Eval `--min-duration`/`--stratify` (`rally_seg_eval.py`) + product `stitching.min_rally_duration`. Finding: single-court over-fire is short-FP-dominated (seg_ratio 1.43?1.04) but **multicourt over-fires 1.58× even on long rallies** (? needs court-localization, Lever C) and strict tolP is gated by over-merge contamination, not just FPs. Sharpened §3 + added §7 row P7.

4. user (2026-06-19T15:52:12.577Z)

fatal: Cannot fast-forward to multiple branches.

5. assistant (2026-06-19T15:52:15.461Z)

The doc is loaded. Note the repo has uncommitted/local state and a pull issue, but I'll proceed read-only on what's local. Now let me verify the cited code for each claim.

6. user (2026-06-19T15:52:15.967Z)

```
"""Pure per-window person-cue features for multi-signal rally fusion (MULTI_SIGNAL_FUSION_PLAN §3.2/§4).

NO I/O, no torch, no cv2 ? mirrors `rally_gate.py`'s discipline so the whole feature
math is unit-testable in a GPU-free dev container. The detection (person boxes,
shuttle trajectory) is produced elsewhere (`PersonDetector`, the WASB runner); this
module only turns those signals into a handful of **scale/translation-invariant**
features (counts / ratios / zone-membership ? never raw pixel coordinates, which would
```

```
memorise this court/camera and fail to generalise off a small labelled set).

Conventions:
- A person box is ``(x1, y1, x2, y2)`` in **pixels**; ``(track_id, box)`` when motion needs it.
- ``per_frame`` is ``{frame_idx: [(track_id, box), ...]}`` keyed by the ORIGINAL video frame
  index (so it aligns directly with the shuttle trajectory's ``.frame`` ? no clip re-timing).
- A shuttle point has ``.frame``, ``.visible``, ``.x``, ``.y`` (pixels) ? the WASB trajectory shape.
- ``court_polygon`` is normalised 0-1 ``[(x, y), ...]`` or ``None`` (None ? no mask, every
  detection counts ? the player-count-only mode).
- ``net`` is ``(axis 'x'|'y', position 0-1 normalised)`` or ``None`` (None ? two-sided = 0.0).

Every function degrades gracefully (returns 0.0 / empty) on missing inputs; outputs are
floats in [0, 1] except ``mean_player_motion`` (a non-negative normalised speed).
"""
from __future__ import annotations

from typing import Dict, List, Optional, Sequence, Tuple

BBox = Tuple[float, float, float, float]
Net = Tuple[str, float]


def point_in_polygon(point: Tuple[float, float], polygon: Sequence[Tuple[float, float]]) -> bool:
    """Ray-casting point-in-polygon. `point` and `polygon` share a coordinate space
    (we use normalised 0-1). Empty/degenerate polygon (<3 vertices) ? False."""
    if polygon is None or len(polygon) < 3:
        return False
    x, y = point
    inside = False
    n = len(polygon)
    j = n - 1
    for i in range(n):
        xi, yi = polygon[i]
        xj, yj = polygon[j]
        # Does the horizontal ray at y cross edge (i, j)?
        if (yi > y) != (yj > y):
            x_cross = (xj - xi) * (y - yi) / (yj - yi) + xi if yj != yi else xi
            if x < x_cross:
                inside = not inside
        j = i
    return inside


def _foot_norm(box: BBox, frame_width: float, frame_height: float) -> Tuple[float, float]:
    """Normalised foot point (bottom-centre) of a person box ? where the player stands,
    the right anchor for court membership. Returns (0,0) if frame dims are unusable."""
    if frame_width <= 0 or frame_height <= 0:
        return (0.0, 0.0)
    x1, _y1, x2, y2 = box
    return ((x1 + x2) / 2.0 / frame_width, y2 / frame_height)


def _center_norm(box: BBox, frame_width: float, frame_height: float) -> Tuple[float, float]:
    if frame_width <= 0 or frame_height <= 0:
        return (0.0, 0.0)
    x1, y1, x2, y2 = box
```

```
    return ((x1 + x2) / 2.0 / frame_width, (y1 + y2) / 2.0 / frame_height)


def person_in_court(box: BBox, frame_width: float, frame_height: float,
                    court_polygon: Optional[Sequence[Tuple[float, float]]]) -> bool:
    """Is this person on the court? Foot-point-in-polygon when a polygon is supplied;
    True (count everyone) when it is None. Unusable frame dims (w/h <= 0) make the foot
    point meaningless, so a masked check returns False rather than testing a bogus (0,0)."""
    if court_polygon is None:
        return True
    if frame_width <= 0 or frame_height <= 0:
        return False
    return point_in_polygon(_foot_norm(box, frame_width, frame_height), court_polygon)


def in_court_counts(per_frame: Dict[int, List[Tuple[int, BBox]]], frame_width: float,
                    frame_height: float,
                    court_polygon: Optional[Sequence[Tuple[float, float]]]) -> List[int]:
    """Per-frame count of in-court persons (one entry per sampled frame)."""
    return [
        sum(1 for _tid, box in dets if person_in_court(box, frame_width, frame_height, court_polygon))
        for _frame_idx, dets in sorted(per_frame.items())
    ]


def _median(vals: Sequence[float]) -> float:
    if not vals:
        return 0.0
    s = sorted(vals)
    n = len(s)
    return float(s[n // 2] if n % 2 else 0.5 * (s[n // 2 - 1] + s[n // 2]))


def players_in_court(counts: Sequence[int]) -> float:
    """Median per-frame in-court count (?2 singles / 4 doubles; 0-1 = dead time)."""
    return _median(list(counts))


def in_court_ratio(counts: Sequence[int], min_players: int = 2) -> float:
    """Fraction of sampled frames with >= `min_players` in court (track stability vs flicker)."""
    if not counts:
        return 0.0
    return sum(1 for c in counts if c >= min_players) / len(counts)


def mean_player_motion(per_frame: Dict[int, List[Tuple[int, BBox]]],
                       frame_width: float, frame_height: float) -> float:
    """Mean per-track centre displacement between consecutive sampled frames, with x
    normalised by frame width and y by frame height (per-axis; uniform-scale-invariant,
    not aspect-ratio-isotropic). 0.0 if <2 frames or no shared tracks."""
    if frame_width <= 0 or len(per_frame) < 2:
        return 0.0
    frames = sorted(per_frame.keys())
    steps: List[float] = []
    for a, b in zip(frames, frames[1:]):
        prev = {tid: box for tid, box in per_frame[a]}
        for tid, box in per_frame[b]:
            if tid in prev:
                cx0, cy0 = _center_norm(prev[tid], frame_width, frame_height)
                cx1, cy1 = _center_norm(box, frame_width, frame_height)
                steps.append(((cx1 - cx0) ** 2 + (cy1 - cy0) ** 2) ** 0.5)
```

```python
    if not steps:
        return 0.0
    return sum(steps) / len(steps)


def two_sided_motion(per_frame: Dict[int, List[Tuple[int, BBox]]], frame_width: float,
                     frame_height: float, net: Optional[Net],
                     court_polygon: Optional[Sequence[Tuple[float, float]]]) -> float:
    """Fraction of sampled frames with >=1 in-court person on BOTH net halves.

    A real rally is two-sided; a single player feeding/holding the shuttle is one-sided.
    The net splits the play axis at `net=(axis, position)` in normalised coords. None ?
0.0.
    """
    if net is None or not per_frame:
        return 0.0
    axis, pos = net
    both = 0
    for _frame_idx, dets in per_frame.items():
        near, far = False, False
        for _tid, box in dets:
            if not person_in_court(box, frame_width, frame_height, court_polygon):
                continue
            fx, fy = _foot_norm(box, frame_width, frame_height)
            coord = fx if axis == "x" else fy
            if coord < pos:
                near = True
            else:
                far = True
        if near and far:
            both += 1
    return both / len(per_frame)


def _shuttle_in_box(sx: float, sy: float, box: BBox, frame_width: float, frame_height:
float,
                    pad_frac: float) -> bool:
    """Is the (pixel) shuttle point inside a person box, expanded by `pad_frac` of frame
    width/height (the 'adjacent / in-hand' tolerance)?"""
    x1, y1, x2, y2 = box
    px = pad_frac * frame_width
    py = pad_frac * frame_height
    return (x1 - px) <= sx <= (x2 + px) and (y1 - py) <= sy <= (y2 + py)


def shuttle_player_colocation(per_frame: Dict[int, List[Tuple[int, BBox]]],
shuttle_points,
                              frame_width: float, frame_height: float,
                              pad_frac: float = 0.03) -> float:
    """Fraction of (visible shuttle frames that were also person-sampled) in which the
    shuttle sits inside/adjacent to a person box. **held/in-hand ? high; in-flight ?
~0**
    ? the direct held-shuttle discriminator. 0.0 if no aligned visible shuttle frames.
    """
    if not per_frame or frame_width <= 0 or frame_height <= 0:
        return 0.0
    aligned = 0
    colocated = 0
    for p in shuttle_points or []:
        if not getattr(p, "visible", False):
            continue
        dets = per_frame.get(int(p.frame))
        if dets is None:          # shuttle frame wasn't person-sampled ? can't judge
            continue
        aligned += 1
```

```
186             if any(_shuttle_in_box(p.x, p.y, box, frame_width, frame_height, pad_frac)
187                    for _tid, box in dets):
188                 colocated += 1
189         return colocated / aligned if aligned else 0.0
190
191
192     # The ~5 person features persisted into candidate `stats` (order = the learned-fusion
193     # feature columns appended after the shuttle features in distill_local.FEATURE_NAMES).
194     PERSON_FEATURE_NAMES = [
195         "players_in_court", "two_sided_motion", "mean_player_motion",
196         "in_court_ratio", "shuttle_player_colocation",
197     ]
198
199
200     def compute_person_features(per_frame: Dict[int, List[Tuple[int, BBox]]],
shuttle_points,
201                                 frame_width: float, frame_height: float,
202                                 court_polygon: Optional[Sequence[Tuple[float, float]]] =
None,
203                                 net: Optional[Net] = None,
204                                 min_players: int = 2) -> Dict[str, float]:
205         """Aggregate the per-window person features from per-frame detections + the in-
window
206         shuttle trajectory. Pure; returns exactly `PERSON_FEATURE_NAMES`, all floats."""
207         counts = in_court_counts(per_frame, frame_width, frame_height, court_polygon)
208         return {
209             "players_in_court": players_in_court(counts),
210             "two_sided_motion": two_sided_motion(per_frame, frame_width, frame_height, net,
court_polygon),
211             "mean_player_motion": mean_player_motion(per_frame, frame_width, frame_height),
212             "in_court_ratio": in_court_ratio(counts, min_players),
213             "shuttle_player_colocation": shuttle_player_colocation(
214                 per_frame, shuttle_points, frame_width, frame_height),
215         }
216
```

7. user (2026-06-19T15:52:17.327Z)

EXISTS

8. assistant (2026-06-19T15:52:24.332Z)

`shuttle_player_colocation` is indeed BUILT in `fusion_features.py` (function at line 168, in `PERSON_FEATURE_NAMES` line 197, computed by `compute_person_features` line 213). Now let me verify it's wired in `fusion_hybrid.py` and check the manifest count.

9. user (2026-06-19T15:52:24.932Z)

```
list
len: 15
[{'name': 'adarsh_avi_singles', 'trajectory': 'C:/Users/avidu/Projects/Annotation
Setup/Collect/Trajectories/adarsh_avi_full_traj.csv', 'fps': 59.94, 'frame_width': 1920,
'labels': 'C:/Users/avidu/Projects/Annotation Setup/Collect/Golden Labelled/Adarsh and
Avi.rallies.csv'}, {'name': 'kushagra_singles', 'trajectory':
'C:/Users/avidu/Projects/Annotation Setup/Collect/Trajectories/kushagra_traj.csv', 'fps': 59.94,
'frame_width': 1920, 'labels': 'C:/Users/avidu/Projects/Annotation Setup/Collect/Golden
Labelled/KushagraYashAviFirstVideo.rallies.csv'}, {'name': 'largetest_doubles', 'trajectory':
'C:/Users/avidu/Projects/Annotation Setup/Collect/Trajectories/largetestvideo_traj.csv', 'fps':
29.97, 'frame_width': 1920, 'labels': 'C:/Users/avidu/Projects/Annotation Setup/Collect/Golden
Labelled/LargeTestVideo.rallies.csv'}]
```

10. user (2026-06-19T15:52:25.124Z)

backend\pipeline\segmenters\fusion_hybrid.py

11. assistant (2026-06-19T15:52:27.382Z)

The manifest has 15 entries. Now let me verify the `fusion_hybrid.py` wiring referenced in the proposed fixes (`cue_flags.person`, `_person_features`).

12. user (2026-06-19T15:52:29.068Z)

```
10:  is explicitly enabled (``indexing.fusion.cue_flags.person``), the person-cue features
11:  (`fusion_features`), persisting them into ``candidate_segments.stats`` for the learned
36:from backend.pipeline.detectors import fusion_features as ff
88:        if fcfg.get("cue_flags", {}).get("person", False):
89:            stats.update(self._person_features(cw, points, fps, frame_width, video_path, fcfg))
93:    # Person features (from _person_features when cue_flags.person) are now persisted for the harness
111:    def _person_features(self, cw: Dict[str, Any], points: List[Any], fps: float,
132:            return {k: 0.0 for k in ff.PERSON_FEATURE_NAMES}
138:        return ff.compute_person_features(det["frames"], shuttle_in, fw, fh,
160:        person_on = bool(fcfg.get("cue_flags", {}).get("person", False))
```

13. assistant (2026-06-19T15:52:43.795Z)

The wiring is confirmed: `fusion_hybrid.py:88` gates on `cue_flags.person`, calls `_person_features` (line 89/111), which calls `ff.compute_person_features` (line 138), which produces `shuttle_player_colocation`. It's built but default-OFF (gated on a flag that defaults to `False`).

Now I have all the evidence I need. Let me verify each proposed fix's stale_text is a verbatim unique match and assess corrections.

Fix 1 ? stale_text: `  - *Shuttle-in-hand / out-of-play* ? \`shuttle_player_colocation\` (shuttle box ? person box = held) \`[design]\` from the fusion plan §4,` ? This appears verbatim on line 56. The `[design]` tag IS stale: the feature is built (`fusion_features.py:168`). Per the doc's honesty key (line 7), `[design]` = "proposed". The corrected_text adds `[verified]` + accurate built/default-OFF/unmeasured qualifier. This is accurate but I'll tighten ? the corrected text says "built in `fusion_features.py`" which matches reality. Good. CONFIRM.

Fix 2 ? stale_text: `person producer in \`yolo_hybrid\` (built) · \`shuttle_player_colocation\` (new feature) · \`scratch/ingest_golden.py\`` ? appears verbatim on line 61. "(new feature)" in a "what's new vs there" list implies not-yet-built, but it IS built. The correction is accurate. CONFIRM.

Fix 3 ? stale_text: `**Caveat:** two golden clips so far (n=13 corpus, GX010141+GX010137) ? directional, not conclusive.` ? appears verbatim on line 12. Manifest has 15 entries. n=13 ? n=15. The "two golden clips so far" (n=2 ground-truth-scored) is preserved. CONFIRM.

Fix 4 ? stale_text references line 64: `the corpus is n=12`. Let me confirm the full verbatim string.

The line 64 reads: `- **n is small.** One golden clip scored so far (GX010141); the corpus is n=12 with several weak clips. Per-video conclusions firm up only as the owner labels more (GX010137 / GX020140 / a large-count clip next).` ? matches the proposed stale_text verbatim. n=12 ? n=15 is correct per manifest. CONFIRM.

Fix 5 ? stale_text line 26: `| D3 | **Golden labels = ground truth; Gemini = reference, not truth** | n=12 corpus | Gemini under-detects too ? agreement ? accuracy. Report both, gate on golden. |` ? matches verbatim. n=12 ? n=15. CONFIRM.

Let me double-check there's consistency on the corpus-size figures across the doc ? there are several `n=12`/`n=13`/`n=15` references, and I want to ensure the manifest is the right ground-truth source for "corpus" size and that my matches are unique.

14. user (2026-06-19T15:52:45.441Z)

12:[Omitted long matching line]
26:| D3 | **Golden labels = ground truth; Gemini = reference, not truth** | n=12 corpus | Gemini under-detects too ? agreement ? accuracy. Report both, gate on golden. |
40:- **Corpus aggregate (n=13, milestone):** tIoU F1@0.5 **0.474** [0.359, 0.579], F1@0.25 0.633; R2 tolF1 0.359. GX010137 (0.822) is the corpus's **best** clip; GX010141 (0.510) is above the mean. `[verified]`
48:[Omitted long matching line]
64:- **n is small.** One golden clip scored so far (GX010141); the corpus is n=12 with several weak clips. Per-video conclusions firm up only as the owner labels more (GX010137 / GX020140 / a large-count clip next).
81:| P7 | **Long-rally (>5s) quality lens** (eval `--min-duration`/`--stratify`) + reel knob (`stitching.min_rally_duration`), both default-OFF. **Measured n=15:** single-court over-fire is short-FP-dominated (seg_ratio 1.43?1.04) but **multicourt over-fires 1.58× even on long rallies** (? court-localization, Lever C) and strict tolP doesn't rise (over-merge-contaminated). See [`QUALITY_ITERATIONS.md`](QUALITY_ITERATIONS.md) "P7". | ? | No (eval+cfg, default-OFF) | ? measured | this PR |
104:- 2026-06-18 ? **P7 long-rally (>5s) quality lens + reel knob (both default-OFF), measured n=15.** Eval `--min-duration`/`--stratify` (`rally_seg_eval.py`) + product `stitching.min_rally_duration`. Finding: single-court over-fire is short-FP-dominated (seg_ratio 1.43?1.04) but **multicourt over-fires 1.58× even on long rallies** (? needs court-localization, Lever C) and strict tolP is gated by over-merge contamination, not just FPs. Sharpened §3 + added §7 row P7.

15. assistant (2026-06-19T15:52:53.580Z)

I need to be careful here. There's a subtlety with the corpus-size figures. The doc has multiple distinct numbers:
- Line 12, 26, 64: `n=12`/`n=13` referring to "corpus"
- Line 40: `Corpus aggregate (n=13, milestone)` ? a measured aggregate scored over a specific corpus
- Lines 81, 104: `n=15` for the P7 long-rally measurement

The proposed fixes update `n=12`/`n=13` ? `n=15` based on the manifest having 15 entries. But I must verify the manifest is genuinely "the corpus" referenced. The P7 measurement (line 81) already uses `n=15` and was the most recent work. Let me confirm the manifest is what these "corpus" figures track, and check whether line 40's `n=13` (the measured aggregate `0.474 [0.359, 0.579]`) is a different, frozen measurement that should NOT be bumped.

The proposed fixes deliberately do NOT touch line 40 (the measured aggregate with CI) ? that's a specific measurement frozen at its n. Good. But I need to confirm that the "corpus" the prose refers to (lines 12, 26, 64) is indeed the manifest, and that 15 is the right current number. Let me check the manifest names and the P7/QUALITY_ITERATIONS provenance for "n=15".

16. user (2026-06-19T15:52:55.289Z)

adarsh_avi_singles
kushagra_singles
largetest_doubles
gbaaddy
testlarge_short
mahadevpura_singles
mahadevpura_2
GX010128
mahadevpura_1
GX030094
Badminton_BXH_2
GX010141
GX010137
Boxhill_Doubles
GX020094

17. user (2026-06-19T15:52:56.543Z)

---STATUS---

18. assistant (2026-06-19T15:53:11.161Z)

The manifest is committed (not dirty), has 15 entries, and includes both GX010141 and GX010137. The P7 work (line 81) measured `n=15` against this same corpus, confirming 15 is the current corpus size.

Now let me cross-check that the doc's own prose treats "the corpus" as this LOVO manifest. The P7 row (line 81) and changelog (line 104) both cite `n=15` as the current measured corpus, while lines 12/26/64 say `n=12`/`n=13` ? an internal inconsistency within the same document that confirms the lower numbers are stale relative to the latest P7 measurement. This is strong evidence.

Let me verify one more time that line 40's `n=13` aggregate is genuinely a frozen measurement that the fixes correctly leave alone (it carries a specific CI `[0.359, 0.579]`), versus the prose "corpus" figures. The proposed fixes do not touch line 40 ? consistent with it being a frozen measured value. I'll note this is the correct scoping.

All five fixes verified. Let me confirm each stale_text is a unique substring (no duplicates).

- Fix 1 (line 56): the `*Shuttle-in-hand / out-of-play*` phrase appears only once. Unique.
- Fix 2 (line 61): `person producer in \`yolo_hybrid\` (built)` appears only once. Unique.
- Fix 3 (line 12): `**Caveat:** two golden clips so far (n=13 corpus` appears only once. Unique.
- Fix 4 (line 64): `**n is small.** One golden clip scored so far` appears only once. Unique.
- Fix 5 (line 26): the D3 row appears only once. Unique.

All confirmed. For the corpus-size fixes (3, 4, 5), I'll mark them `revise` only if I'd change the correction; the proposed corrections are accurate and minimal, so they are `confirm`. For fixes 1 and 2, the `[design]`?built corrections are accurate but I'll tighten fix 1's corrected_text slightly to preserve the doc's exact tag style ? the proposed correction is faithful, so `confirm`.

Let me do a final adversarial check on whether the manifest count is truly "the corpus" or whether some entries are non-golden (e.g., the corpus aggregate at line 40 used n=13). The fact that two entries ? `GX010137` and `GX010141` ? are the ground-truth-scored ones (n=2), while the full LOVO manifest is the broader corpus, means corpus=15 is now correct and the doc's older 12/13 figures are stale. This matches the doc's own P7 `n=15`.

All five fixes survive. Submitting.

19. user (2026-06-19T15:53:34.629Z)

Structured output provided successfully

20. assistant (2026-06-19T15:53:43.093Z)

All five proposed fixes survived adversarial verification.

Verified evidence:
- `shuttle_player_colocation` is genuinely BUILT ? `C:\Users\avidu\Projects\badminton-highlight-indexer\backend\pipeline\detectors\fusion_features.py:168` (function), `:197` (`PERSON_FEATURE_NAMES`), `:213` (`compute_person_features`), wired in `C:\Users\avidu\Projects\badminton-highlight-indexer\backend\pipeline\segmenters\fusion_hybrid.py:88` (gated on `cue_flags.person`, default `False`) ? `:111` `_person_features` ? `:138` `compute_person_features`. So the `[design]`/`(new feature)` tags (fixes 1, 2) are stale; corrected to built-but-default-OFF + unmeasured-on-golden (no overclaim).
- Corpus size: `C:\Users\avidu\Projects\badminton-highlight-indexer\output\human_lovo_manifest.json` has 15 entries (committed, clean tree), and the doc's own most-recent P7 work (lines 81, 104) already uses `n=15` ? confirming the `n=12`/`n=13` prose figures (fixes 3, 4, 5) are stale. The frozen measured aggregate at line 40 (`n=13` with CI `[0.359, 0.579]`) was correctly left untouched.

All `stale_text` strings are exact, unique substrings of `C:\Users\avidu\Projects\badminton-highlight-indexer\docs\RALLY_DETECTION_GAP_CLOSURE.md`. All five verdicts are `confirm`.